

main.c

Run

Share

```
1 //BIN PACKING PROGRAM
2 #include <stdio.h>
3 #include <stdlib.h>
4
5 #define MAX_ITEMS 20
6
7 int n;
8 int capacity;
9 int weight[MAX_ITEMS];
10 int min_bins;
11 int bin_load[MAX_ITEMS];
12
13 // Sort in descending order
14 int compare(const void *a, const void *b) {
15     return (*(int*)b - *(int*)a);
16 }
17
18 void get_initial_upper_bound() {
19     int bins_used = 0;
20     int temp_load[MAX_ITEMS] = {0};
21
22     for (int i = 0; i < n; i++) {
23         int placed = 0;
```

Output

Clear

```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
```

main.c		Run	Output
27	placed = 1;		--- Bin Packing: Branch and Bound --- Enter number of items: 8 Enter bin capacity: 9 Enter weight of 8 items: === Session Ended. Please Run the code again ===
28	break;		
29	}		
30	}		
31	if (!placed) {		
32	temp_load[bins_used] = weight[i];		
33	bins_used++;		
34	}		
35	}		
36	min_bins = bins_used;		
37	}		
38			
39	void solve(int item_idx, int bins_used) {		
40	if (bins_used >= min_bins)		
41	return;		
42			
43	if (item_idx == n) {		
44	if (bins_used < min_bins)		
45	min_bins = bins_used;		
46	return;		
47	}		
48			
49	for (int i = 0; i < bins_used; i++) {		
50	if (bin_load[i] + weight[item_idx] <= capacity) {		
51	bin_load[i] += weight[item_idx];		
52	solve(item_idx + 1, bins_used);		

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```
61 }
62 }
63
64 int main() {
65     printf("--- Bin Packing: Branch and Bound ---\n");
66
67     printf("Enter number of items: ");
68     scanf("%d", &n);
69
70     printf("Enter bin capacity: ");
71     scanf("%d", &capacity);
72
73     printf("Enter weight of %d items:\n", n);
74     for (int i = 0; i < n; i++) {
75         scanf("%d", &weight[i]);
76     }
77
78     // Sort items in descending order (important optimization)
79     qsort(weight, n, sizeof(int), compare);
80
81     get_initial_upper_bound();
82
83     for (int i = 0; i < n; i++)
84         bin_load[i] = 0;
85
86     solve(0, 0);
87 }
```

Output

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```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
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printf("Enter number of items: ");

scanf("%d", &n);

printf("Enter bin capacity: ");

scanf("%d", &capacity);

printf("Enter weight of %d items:\n", n);

for (int i = 0; i < n; i++) {

scanf("%d", &weight[i]);

}

// Sort items in descending order (important optimization)

qsort(weight, n, sizeof(int), compare);

get_initial_upper_bound();

for (int i = 0; i < n; i++)

bin_load[i] = 0;

solve(0, 0);

printf("\nMinimum bins required: %d\n", min_bins);

return 0;

}

Output

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--- Bin Packing: Branch and Bound ---

Enter number of items: 8

Enter bin capacity: 9

Enter weight of 8 items:

=== Session Ended. Please Run the code again ===